MILITARY SPECIFICATION SHEET
ELECTRON TUBE, CATHODE RAY CHARGE STORAGE
TYPE 8602/CK1383A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Electronic output, dual-gun recording charge storage tube, magnetic deflection and magnetic focus. See figure 3

ABSOLUTE-MAXIMUM RATINGS: Writing end (voltages with respect to writing gun cathode):

Parameter: (see note 1)	EWf	EWc1	EWc2	EWb1	EWL1	EWDec	EWhk	RG1	EK
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Ohms	V
Maximum:	6.6	0	800	5000	1000	1000	10	500K	
Minimum:	6.6	-100	--	--	--	--	-125	--	
Test Conditions: (see note 7)	6.3	- 20 - 50 Adjust	450	4500	400 600 Adjust	800 925 Adjust	--	100K	0

ABSOLUTE-MAXIMUM RATINGS: Reading end (voltage with respect to reading gun cathode except where noted):

Parameter: (see note 1)	ERk	ERf[1/]	ERc1	ERc2	ERb	ERL1	ERL2	ERDec	Ess	Ecoll	ERhk	Rg1
Unit:	V	V	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	Ohms
Maximum:	800	6.6	0	800	5000	1500	1000	1500	500	1000	10	500K
Minimum:	--	6.0	-300	--	--	--	--	--	--	--	-125	--
Test Conditions: (see note 7)	400	6.3	-100 -250 Adjust	600	4100	1300 700 Adjust	250 600 Adjust	1300 700 Adjust	0 10 Adjust	500 - 40 Adjust	--	100K

ABSOLUTE-MAXIMUM RATINGS: Storage assemble

Parameter: (see note 1)	Esa[2/]
Unit:	V
Maximum:	1600
Minimum:	--

Test Conditions: --

1/ Voltage with respect to writing gun cathode.

2/ Voltage between adjacent screens.

8602/CK1383A
FSC 5960

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GENERAL:

Qualification - Required (see note 23) Marking - (see note 22)

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Qualification inspection					
1026	Temperature range, operating					
	Low temperature	warm-up, time =3 min.	----	-65	---	°C
	High temperature	warm-up, time =1 min.	----	+55	---	°C
1216	Base material insulating quality	zone 5 min.	----	---	---	---
1101	Secureness of base, base insert, and cgs	----	----	---	---	---
5111	Vibration	----	----	---	---	---
1042	Shock	----	----	---	---	---
	Quality conformance inspection, part 1					
	Static tests (see note 2)					
5201	Electrode current and voltage breakdown (magnetic types)	----	----	---	---	---
5251	Electrode leakage currents (Grid No. 1)	----	----	---	3	uA
5251	Electrode leakage current (Grid No. 2)	----	----	---	5	uA
5251	Electrode leakage current (Anode No. 1)	----	----	---	5	uA
5402	Grid cutoff voltage (reading gun)	See note 4	ERco	-100	-260	V
5402	Grid cutoff voltage (writing gun)	See note 4	EWco	-20	- 50	V
1256	Electrode current (reading gun)	ERb = 2KV; ERc1 = 0	IRkk	2000	---	uA
1256	Electrode current (writing gun)	EWc1 = 0	IWk	400	---	uA
----	Electrode current (Beam) (reading gun)	See note 3	IRb	100	---	uA

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Quality conformance inspection, part 1 (Cont'd)					
1256	Electrode current (Beam) (writing gun)	EWc1 = 0	IWb	15	---	uA
5403	Modulation (reading gun)	See note 5	ΔERc1	---	30	V
5403	Modulation (writing gun)	See note 6	ΔEWc1	---	15	V
----	Dynamic test "A"	See note 7 R. F. Separation EWDec. = 800Vdc ERL1 = 1300 Vdc	---- ERL2	--- 300	--- 600	--- Vdc
----	Output current	See note 9	Io	2	---	ua
----	Long storage	See note 10 Io = 2 ua	----	180	---	sec.
----	Output signal uniformity	See note 11				
	-Quality area signal-to-shading ratio		----	5:1	---	ratio
	-Signal-to-disturbance ratio		----	8:1	---	ratio
	-Signal-to-background shading ratio		----	10:1	---	ratio
	-Signal-to-background disturbance ratio		----	10:1	---	ratio
5231	*Spot position*	*Reading beam (see notes 2 and 8)*	----	---	4	%
		Writing beam (see notes 2 and 8)	----	---	4	%
----	Storage surface blemishes	See note 12	----	---	---	---
----	Fast erase	See note 13	----	---	2	sec.
----	Spurious signal build-up					
	- I on spot	See note 14	----	---	10	%
	- Grid emission	See note 15	----	---	10	%
----	Resolution	See note 16 Center Edge	---- ----	1100 950	--- ---	TV lines TV lines
----	Dynamic test "B"	See note 7 Write Video cancellation	EWDec	875	925	Vdc
		ERL1 = 700 Vdc	----	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITION	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Quality conformance inspection, part 1 (cont'd)					
			ERL1	250	450	Vdc
----	Output current	See note 9	Io	2	---	ua
----	Long storage	See note 10 Io = 2 ua	----	180	---	sec
----	Output signal uniformity	See note 11				
	-Quality area signal-to-shading ratio	----	---	5:1	---	ratio
	-Signal-to-disturbance ratio	----	---	8:1	---	ratio
	-Signal-to-background shading ratio	----	---	10:1	---	ratio
	-Signal-to-background disturbance ratio	---	---	10:1	---	ratio
	Quality conformance inspection, part 2					
1331	Direct interelectrode capacitance	Collector screen to all other screens and external shielding	C	20	35	p^f
		Writing grid 1 to all other writing gun elements	C	10	16	p^f
		Reading grid 1 to all other reading gun elements and shielding (see note 17)	C	17	23	p^f
		Change in reading grid 1 (see note 18)	AC	---	5	p^f
1301	Heater or filament current (reading gun)	----	IRf	540	660	mA
1301	Heater or filament current (writing gun)	----	IWf	540	660	mA
---	Dynamic range	See note 19	---	5	---	steps

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN.	LIMITS MAX.	UNITS
	Quality conformance inspection, part 3					
----	Life test provisions	Group D (see note 20)	----	1000	---	hrs.
----	Life test end points	Center resolution (see not 21)	----	900	---	lines
----	Additional life test end point for grid emission	Scan as for grid emission (see note 15) test but for 40 sec.	----	---	10	%

NOTES:

1. The following additional symbols have been used:

EW or IW	=	Writing gun voltage or current of indicated element
ER or IR	=	Reading gun voltage or current of indicated element
EL1	=	Voltage on collimating lens 1
EL2	=	Voltage on collimating lens 2
EDecel.	=	Voltage on decelerator screen
E St. Scr.	=	Voltage on storage screen
E Coll.	=	Voltage on collector screen
Esa	=	Voltage on adjacent screens

2. To avoid damage to the storage assembly, deflection shall be provided for the reading and writing beams whenever either gun of the tube is providing beam current, except during the focus-coil alignment procedure and beam-positioning tests. When an undeflected beam is used, precautions shall be taken to limit the sharpness of focus and the beam intensity to levels that shall not cause screen damage. For beam and cathode current tests, the beam of the gun under test shall be unfocused and deflected so that the storage surface is over-scanned in all directions. A gun shall not be left in the zero-bias condition for more than 30 seconds.

3. The storage screen, collector screen, reading and writing decelerator screens, and collimating lenses shall be connected in parallel. The voltage applied to these elements shall be 4545 Vdc with respect to the cathode of the gun under test. All voltages for the gun under test shall be at their specified test levels, and the opposite gun shall be cut off. The current measured from the storage assembly and lenses, thus connected in parallel, shall be total beam current (Ib) for the gun under test.

4. Grid cutoff voltage is here defined as a grid 1 potential such that the corresponding total beam current is less than or equal to 0.05 microampere and greater thab 0.00 microampere.

5. The storage assembly and lenses shall be connected as specified in note 3. The other reading gun voltage shall be at their specified test levels. With reading grid 1 voltage set initially at grid cutoff voltage, the change in reading grid 1 potential to produce a reading beam current of 10 microamperes shall not exceed the limit specified.

6. The storage assembly and lenses shall be connected as specified in note 3. The other writing gun voltages shall be at their specified test levels. With the writing grid 1 voltage set initially at grid cutoff voltage, the change in writing grid 1 potential to produce a writing beam current of 5 microamperes shall not exceed the limit specified herein.

7. Dynamic Tests - The storage tube output for these tests shall be monitored visually on a television display as well as electrically with an oscilloscope ("A" scope) displaying the readout of lines in the output scan.

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a. Equipment - 525-line 2:1 interlaced television rasters shall be employed for both reading and writing scans. The deflection drives for both reading and writing scans shall be linear to 1.0 percent or better. An interlace field-blanking capability shall be available for the writing raster. The "A" scope shall be Model 545 Tektronix oscilloscope or equivalent. The dynamic tests to follow shall be made with the tube operating in the continuous reading, writing, and automatic prime mode of operation and with external magnetic components properly aligned.

Two methods of eliminating the undesired output due to the interception of part of the writing beam by the collector shall be used during test; R. F. separation and write video cancellation.

a. R. F. Separation - Radio Frequency modulation of the reading beam shall be used to provide separation of reading and writing information. Modulation of the reading beam shall be accomplished by applying a 30-megahertz signal to the grid 1 of the reading gun. (A minimum of 30 volts, peak to peak, of 30-megahertz drive shall be available at the reading grid 1.) Readout of the storage tube shall be accomplished by means of an rf amplification and demodulation system having a response such that its 3-db bandwidth is equal to or greater than 12 megahertz.

b. Write Video Cancellation - Secondary emission for the write side of the collector screen shall be used to cancel the collector output current due to interception of writing beam current. The voltage difference between the write decelerator and the collector shall be adjusted to obtain minimum write video signal in the collector output signal. A video amplifier having at least 6 megahertz bandwidth shall be used to amplify the collector output signal.

8. The distance between the centers of the undeflected, unfocused reading and writing spots and the geometric center of the storage surface shall be accurately measured and expressed as a percentage of the storage surface diameter. This percentage shall not exceed the limit specified herein.

9. With the reading and writing rasters reduced to a rectangle having a 4:3 aspect ratio inscribed in the quality area of the storage surface (defined as a circle whose diameter is 90 percent of the storage surface diameter and whose center is the geometric center of the storage surface), with the collector voltage set at 500 volts with respect to reading cathode, adjust writing beam current and writing time to obtain 90 percent of the output level for a saturated write (saturation being the point beyond which an increase in writing beam current provides no increase in output signal amplitude). Reading and writing beams shall be focused so that best focus is obtained at the center of the storage area. Adjust storage-screen voltage for the best compromise between output signal amplitude and uniformity. Measure the output current amplitude employing the "A" scope presentation displaying horizontal lines in the reading scan in the focused portion of the storage area. The readout amplification system shall be calibrated to convert voltage (as measured by the "A" scope at the readout amplifier output) to storage tube output current.

10. Under the operating conditions specified for note 9, the storage surface will be written to 90 percent of the output level for a saturated write and the writing beam shall be cut off. The long storage capability of the tube shall be such that a minimum of 50 percent of the originally-stored signal level remains 180 seconds after the writing beam is cut off.

11. a. Definitions - The terms shading, disturbance, signal-to-shading ratio, signal-to-background-shading ratio, signal-to-disturbance ratio, and signal-to-background-disturbance ratio shall be used to specify output signal uniformity of the storage tube. Definitions of these terms shall be in accordance with those in IRE Standards on Electron Tubes - Methods of Testing - 621RE7.S1. All uniformity measurements shall be made within the quality area. (See note 9 for definition of quality area.)

Storage screen defects smaller than 0.5 percent of storage screen diameter shall be measured to determine if they shall be classified as blemishes (See note 12). Defects larger than 0.5 percent of storage screem diameter shall be included in the disturbance test.

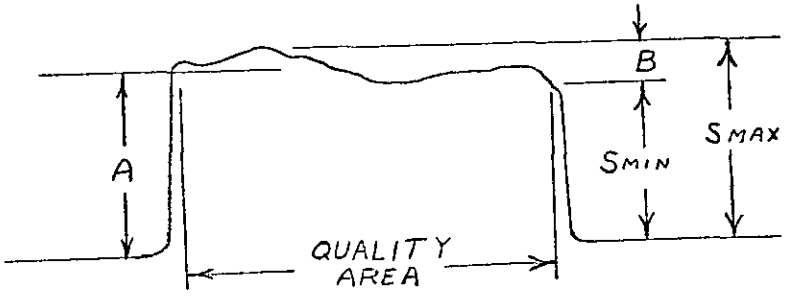
b. Test and Test Conditions - For all Output Signal Uniformity Tests the tube shall be operated with both scans adjusted so the entire storage area is inscribed in the reading and writing rasters and the writing beam unfocused to provide uniform coverage of the storage surface by the lines of the writing beam.

The "A" scope presentation shall be adjusted to maje measurements along the single lines of the readout scan which exhibit the lowest ratios within the quality area.

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The signal-to-shading and signal-to-disturbance tests shall be conducted with the writing beam current adjusted to produce an average output signal amplitude of approximately 90 percent of saturation.

The signal-to-background-shading and signal-to-background-disturbance tests shall be conducted with the writing beam cut off. The measurement shall be made after a normal erase but with the prime storage screen voltage the same as used for read. The average signal amplitude used in determining the ratio shall be obtained by writing to 90 percent of saturation.



$A = \frac{S_{MAX} + S_{MIN}}{2}$ $\qquad B = S_{MAX} - S_{MIN}$

$\text{SIGNAL TO SHADING RATIO} = \frac{A}{B}$

Figure 1. Signal to shading ratio



C = DISTURBANCE VARIATION

$\text{SIGNAL TO DISTURBANCE RATIO} = \frac{A}{C}$

Figure 2. Signal to disturbance ratio

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12. With both scans adjusted so that the entire storage area is inscribed in the reading and writing rasters, and with the writing beam unfocused to provide uniform coverage of the storage surface by the lines of the writing raster, the storage surface appear on the monitor as a circular white are and defects appear as bright and dark spots. Writing beam current shall be adjusted so that the storage tube output signal is slightly below its saturated level. Reading focus shall be adjusted for best resolution, reading scan size shall be adjusted for magnification of small defects, and storage time shall be adjusted by collector voltage setting for decay of output signal by 40 percent to 60 percent in four seconds in the particular area of the storage surface being investigated for blemishes.

Defect amplitude shall be defined in terms of the change in amplitude of the storage tube output signal resulting for a defect and expressed as a percentage of the reference white signal obtained in unaffected areas adjacent to the defect. Defect amplitude measurements shall be made employing the "A" scope presentation. Blemish minimum-amplitude shall be defined as the minimum change in storage tube output signal due to a defect necessary to classify that defect as a blemish. Blemish size shall be defined in terms of the average diameter expressed as a percentage of the diameter of the storage surface as seen on the TV monitor. Blemish size measurements shall be made employing calibrated overlay gauges.

The following table shall specify the limits for blemishes within the quality area. (See note 9 for definition of quality area.)

BLEMISH TABLE

CLASS	SIZE%	POLARITY	MINIMUM AMPLITUDE	MAXIMUM NUMBER
A	.125 – .250	Bright	10.0%	15
B	.250 – .50	Bright and dark	10.0%	7
		TOTAL A + B		15
		TOTAL A + B in 10% cluster*		3

* A cluster where the diameter of a circle enclosing the cluster is equal to or less than 10 percent of the storage surface diameter.

A defect larger than 0.5 percent of the storage screen diameter shall be considered as a disturbance (See note 11).

13. The tube shall be operated under the conditions specified for note 10. Immediately after the writing beam is cut off, an erase with electrode voltage-switching, sequence, and timing as shown in the chart shall be performed.

STEP	ELECTRODE	VOLTAGE
#1 0.75 sec.	Collector	-25 to -50 Vdc (with respect to reading cathode)
	Storage Screen	+400 Vdc (with respect to reading cathode)
	Reading Grid I	Adjustable (for erasing beam intensity)
#2 1.25 sec.	Collector	-25 to -50 Vdc (with respect to reading cathode)
	Storage Screen	Adjustable (0 - 10 Vdc with respect to reading cathode)
	Reading Grid 1	Adjustable (for priming beam intensity)
#3		All voltages returned to normal reading levels.

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Complete erasure shall be accomplished within the time limits specified. There shall be no evidence of reapperarance of previously-written information in excess of the allowable background shading and disturbance levels (note 11) after five minutes.

14. Spurious signal buildup shall include any change in background level due to ion erasure or any spurious signal, localized or general in nature, anywhere on the storage surface. This buildup shall not exceed the specified percentage of saturated output signal.

Operating conditions should be as follows:

Writing and reading raster size as specified in Note 10;

Writing focus set for focused writing beam;

Writing beam cut off;

Storage tube continuously read-out under long storage conditions as specified in note 10;

Make measurement 240 seconds after erase.

15. With the test conditions as specified in note 15, except with write deflection removed, buildup during 60 seconds shall not exceed the specified percentage of saturated output signal. Storage surface blemishes shall be disregarded in this test.

16. With reading and writing beam currents set to produce an output signal level of 2.0 uA under the conditions specified for "Output Signal Amplitude" tests (note 9) and the collimating lens voltages set at the values for "Output Signal Uniformity" tests (note 11), resolution shall be measured as follows:

The tube shall be operated under conditions of continuous reading, writing and automatic prime. The interlace field of the writing raster shall be blanked and writing scan adjusted so that a series of equally and widely spaced white lines are written. Reading and writing scans shall be orthogonal. Reading and writing focus will be optimized for best resolution at the center of the storage surface. A reference signal level corresponding to the peak-to-peak value of the widely spaced white lines shall be determined on the "A" scope adjusted to present the readout of the horizontal line in the reading scan which is the diameter of the storage surface. The writing raster line-spacing shall then be reduced until the peak-to-peak signal at the center of the storage surface is 50% of the reference amplitude. The TV lone resolution shall be computed as follows:

$$\text{Number of Lines Written} \times 2 \times \frac{\text{Diameter of Storage Screen}}{\text{Width of Written Line Pattern}} = \text{TV lines}$$

This measurement shall be repeated at two points along the diameter which are 90 percent of the distance from the center of the storage surface. Reading and writing focus shall be adjusted for best resolution at each of these points. Resolution for these tests shall not be less than the limit specified.

17. The capacitance of the reading grid 1 to all other elements of the reading gun and reading neck shield, under thermal equilibrium conditions and with heater power applied (ERf = 6.3V), shall be within the limits specified.

18. The change in capacitance of the reading grid 1 to all other elements of the storage tube from initial condition of no heater power applied (ERf = OV), to heater power applied (ERf = 6.3V) and thermal equilibrium attained, shall not exceed the limits specified.

19. Under the conditions of operation specified in note 9, the storage tube shall be capable of storing and reading out onto the TV monitor the test pattern from a 2F21 monoscope or equivalent signal source showing a gray scale wedge with a minimum of five distinguishable steps.

20. Operable rejects from current production lots may be used for life tests. Rejects for such items as mechanical defects and screen defects wherein the overall electrical characteristics of the tube are not affected shall be considered operable rejects acceptable for use in life testing.

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For life testing, the storage tube shall be operated under conditions of continuous reading, writing and storing. Reading and writing scans shall be adjusted so that the storage surface is inscribed within the reading and writing rasters. Operating conditions for the life test shall be as specified under Test Conditions.

21. Tests made to determine life test end point shall be resolution as in Note 16 but with a minimum value as specified for this test, and as specified in "Life test end points, cathode-ray tubes" except for "spray emission," which does not apply to electronic output storage tube testing.

22. In addition to normal markings (Vendor identification, etc), the storage assembly area shall be marked such that individual connections to the assembly are clearly defined; that is, bulb connectors to the individual elements shall be identified.

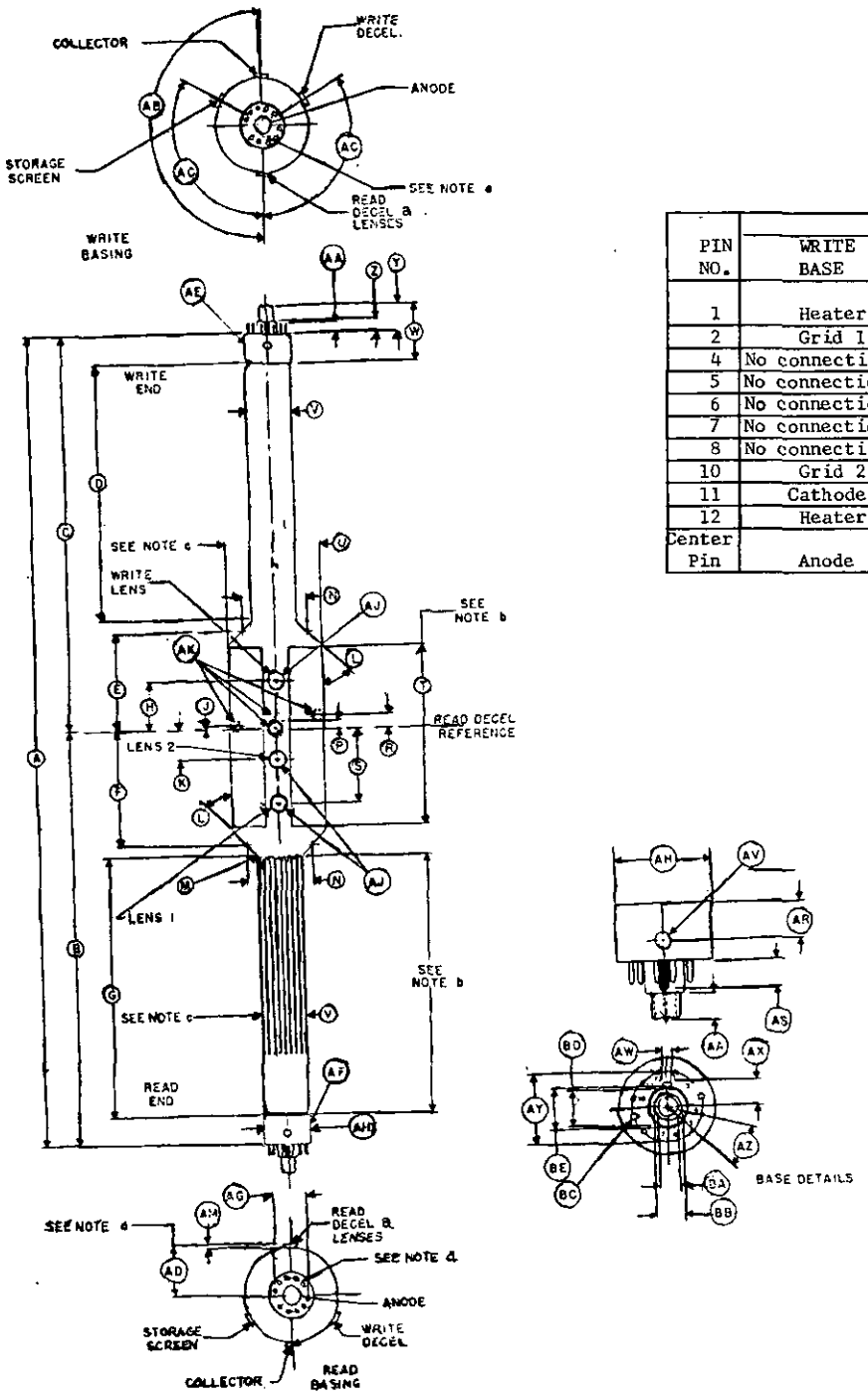
23. The activity responsible for the qualified products list is the Naval Electronic Systems Command (Code 0517), Department of the Navy, Washington D. C. 20360, and information pertaining to qualifications of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification, SO-6" maybe obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pa. 19120.)

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC
(Project 5960-N793)

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PIN NO.	ELEMENT	
	WRITE BASE	PEAD BASE
1	Heater	Heater
2	Grid 1	Grid 1
4	No connection	No connection
5	No connection	No connection
6	No connection	No connection
7	No connection	No connection
8	No connection	No connection
10	Grid 2	Grid 2
11	Cathode	Cathode
12	Heater	Heater
Center Pin	Anode	Anode

Figure 3. Outline drawing for electron tube type 8602/CK1383A.

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Dim.	LIMITS Min.	Max.
A	22.375	22.875
B	11.750 nom.	
C	10.875 nom.	
D	6.500	7.000
E	2.677	3.051
F	3.344	3.718
G	6.750	7.250
H	1.407	1.530
J	.082	.112
K	.844	.968
L	43°	47°
M		.150 R.
N	1.999	2.001
P	.254	.284
R	.427	.457
S	2.219	2.343
T	5.125	5.375
U	2.730 D.	2.960 D.
V	1.350 D.	1.450 D.
W	1.750 nom.	
Y		.875
Z	.500 nom.	
AA	175°	185°
AB	115°	125°
AD		See Note d

Dim.	LIMITS Min.	Max.
AE	Special high altitude base-brown	
AF	Special high altitude base-green	
AG	1.0525 D.	1.0725 D.
AH	1.445 D.	1.475 D.
AJ	J1-22 Contact (see Note g)	
AK	J1-21 Contact (see Note f and Note 1)	
AL		
AM		.135
AR	.545	.567
AS	.422	.442
AV	.250 dia. hole	
AW	.145	.165
AX	.382 nom.	
AY	1.060 D.	1.066 D.
AZ	30° nom.	
BA	.312 nom. dia.	
BB	.438 nom. dia.	
BC	#6-32 NC-2B x .210 max. deep - .156 min. full thread (2 places).	

Dim.	Reference Min.	Center	Max.
BD	.598 D.	.605 D.	.618 D.
BE	.608 D.	.625 D.	.635 D.

NOTES:

a. Keyway of both bases to be coaxial with lens buttons within $\pm 3°$.

b. Shaded areas denote external shielding. Shielding thickness .002 nom.

c. Dimension applies without external shielding.

d. Radial distance from bulb horizontal axis to top surface of any button: 1.615 max.

e. Nominal dimensions are for information and are not required for inspection purposes.

f. Read and write decelerator, storage screen and collector contacts JEDEC designation JI-21.

g. Lens contact JEDEC designation JI-22.

h. All dimensions in inches unless otherwise specified.

i. Bulb to Neck (OD) : 0.060 TIR.
Neck to Neck (CL) : 0.060 TIR.
Neck to Neck (OD) : 0.060 TIR.

j. Mark storage assembly buttons for identification as follows:

ELEMENT	MARK
WRITE LENS	WL
WRITE DECELERATOR	WD
COLLECTOR	COL
STORAGE SCREEN	SS
READ DECELERATOR	RD
READ LENS NO. 1	RL 1
READ LENS NO. 2	RL 2

Figure 3. Outline drawing for tube type 8602/CK1383A (Cont'd).

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